Sibling points Stage 4: aut=3 on j = 0 curves

Sibling points, Stage 4: order-3 automorphism keys (`aut=3`) on j = 0 curves (results, local L0)

What ran. The frozen panel of `PROTOCOL.md` (commit `4ae92d4a40`, frozen before data). It used `ecbench` built from `b460422af6`, which adds `wwalk ... aut=3`: - Keys are min(x, ωx, ω²x), with $\zeta_3(x, y) = (\omega x, y) = \underline{\kappa}$. - A collision at rotation difference d is the row of E_t = ±[κ^d]E_s. - Each canonicalisation charges two field multiplications, as `rho.cubic` charges its rotations. - With `aut` off, existing sessions replay byte-identically: 5,100 runs.

The panel used j = 0, b = 7 prime-order curves (the secp256k1 shape) at p = 34, 37 and 40 bits, with r from 2^33 to 2^40. Each size had five curves, 6 public targets per curve, and 6 rounds plus 1 warmup. The arms were rho-neg, rho-cubic, rho-neg-aa, tw2cj-lazy, tw2cj-lazy-aut3, sib8-p512 and sib8-p512-aut3. It ran on this session's 4-vCPU VM at L0, under bench record `benchmarks/records/20261111T0348Z-sib-stage4-aut3.json`. All numbers are `[pcc:unverified]`, with no pcc claim filed and no validation gate.

Admission. All 3 specs pass: 3 × 1,470 records verified, `verify --replay 8` reproduced 8 of 8 per spec with 0 problems, and the A/A ratio is exactly 1 [pcc:unverified].

Results: paired GAE ratios over 180 pairs per spec (`ecbench compare`, `results.json`)

Each cell gives the ratio at w = 0 / 0.1 / 1. The bracket is the 95 percent interval at w = 0.

ratio	p 34	p 37	p 40	band order
tw2cj-lazy-aut3 / tw2cj-lazy	0.517 [0.490, 0.546]	0.579 [0.557, 0.602]	0.580 [0.552, 0.609]	aut3 lower at every weight, all sizes
sib8-p512-aut3 / sib8-p512	0.595 [0.571, 0.619]	0.587 [0.558, 0.618]	0.568 [0.536, 0.603]	aut3 lower at every weight, all sizes
sib8-p512-aut3 / rho-cubic	**0.886 [0.808, 0.971]** / 1.052 / 1.806	**0.901 [0.811, 1.000]** / 1.073 / 1.848	**0.929 [0.847, 1.020]** / 1.108 / 1.912	unresolved in the band, all sizes
sib8-p512 / rho-cubic	1.489 / 1.773 / 3.055	1.534 / 1.828 / 3.155	1.635 / 1.950 / 3.367	rho-cubic lower at every weight
tw2cj-lazy-aut3 / rho-cubic	1.671	1.793	1.895	rho-cubic lower at every weight
rho-cubic / rho-neg	0.572	0.542	0.515	rho-cubic lower at every weight

Per curve (`band-*.txt`). sib8-p512-aut3 against rho-cubic is unresolved in the band on 13 of 15 curves; on the other 2 (one at p 37, one at p 40), rho-cubic is lower at every weight.

Bit-ops (`stats_bitops.py`, `bitops-*.json`). sib8-p512-aut3 / tw2cj-lazy is 0.335, 0.356 and 0.348. The bitop/v1 instrument question from Stages 2 and 3 persists: it charges rho-cubic 0.99 to 1.09 times rho-neg, where GAE gives 0.52 to 0.57. So bit-op ratios against rho-cubic (sib8-p512-aut3 / rho-cubic 0.56 to 0.59 in `scaling.json`) are not used here.

Scaling (`scaling_fit.py`, a derived statistic). These are slopes of ln(ratio) on ln r over 90 workloads, with 95 percent cluster-bootstrap intervals: - sib8-p512-aut3 / rho-cubic: +0.012 [-0.020, +0.044] - sib8-p512-aut3 / sib8-p512: -0.012 [-0.030, +0.006] - tw2cj-lazy-aut3 / tw2cj-lazy: +0.029 [+0.011, +0.047], driven by p 34 (0.52 against 0.58 at p 37 and 40)


Stage 4

Against the pre-registered expectations

- **H1 holds.** tw2cj-lazy-aut3 / tw2cj-lazy falls in [0.50, 0.66] at every size: 0.517, 0.579, 0.580.
- **H2 holds.** sib8-p512-aut3 / sib8-p512 falls in [0.50, 0.66] at every size: 0.595, 0.587, 0.568.
- **H3 holds.** At w = 0 the upper ends of the sib8-p512-aut3 / rho-cubic intervals are 0.971, 1.000 and 1.020, all below 1.05. The band order is unresolved at every size, as expected.
- **H4 holds.** The slope of ln(sib8-p512-aut3 / rho-cubic) has an interval containing 0.

What this means for prime-field ECDLP

- **The automorphism gain carries into the index-calculus walk.** Orbit-canonical keys cut the counted cost of both C = 2 walks by about 1/sqrt(3), 0.52 to 0.60 [pcc:unverified], the same factor rho-cubic gets over rho-neg (0.52 to 0.57 here).
- **Against the matched rho for the secp256k1 shape.** Sibling points with aut=3 is 0.89 to 0.93 of rho-cubic at w = 0 [pcc:unverified]. That is the first configuration in this program below its matched rho in GAE at w = 0 on these curves; Stage 3's best was 1.48 to 1.54.
  - In the §5a band it is unresolved at every size: 1.05 to 1.11 at w = 0.1 and 1.8 to 1.9 at w = 1. The sibling mechanism's second points and lifts are native operations that rho does not perform.
  - That is not a win. It moves the w = 0 distance remaining from about 1.5 to about 0.9.
- **Nothing here is an exponent statement.** The ratio to rho-cubic is flat in r over 2^33 to 2^40.
- **Not shown.** No wall-time result (L0). No EKS run. No leaderboard change. No claim about secp256k1 itself. On curves without the order-3 automorphism (a = -3, generic) aut=3 does not apply; there the Stage 3 distances stand.

Next

1. Validation gate (replay certificate, validator report, VALIDATION.json) for sib8-p512-aut3 vs rho-cubic; an EKS L2 re-run once the route works.
2. Reconcile bitop/v1 against GAE for rho-cubic, which decides whether the bit-op ratio of about 0.58 means anything.
3. Price the sibling second points more cheaply, or replace them with a native-free variant. The band is decided at w = 1, where they cost the most.

Files: `PROTOCOL.md`, `frozen.sha256`, `spec-j0b7-{34,37,40}.json`, `run_local.sh`, `sessions/` (records gzip-compressed, raw SHA-256 in `sessions/raw.sha256`), `verify-*.json`, `band-*.{json,txt}`, `results.json`, `collect.py`, `scaling.json`, `scaling_fit.py`, `stats_bitops.py`, `bitops-*.json`, `plot.py`, `file:///home/user/crypto/research/curve25519_autolab_20261104/siblings/stage4-aut3/figures/`, `REPORT.pdf`.